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Unit – 1 : Session – 10 : SLO – 1 | NOR gate and universal implementation

■ Part B – Multiple Choice Questions

1. B — $(A + B)'$
2. C — It can implement AND, OR, NOT gates
3. B — Connect inputs together
4. B — $(A' + B)'$
5. D — All of the above

■ Part A – Match the Following

1. NOR gate — c) Output is 1 only when all inputs are 0
2. NOR as NOT — a) Connect both inputs together
3. NOR as OR — NOR the inputs, then invert the output with another NOR: $((A+B))' = A + B$
4. NOR as AND — e) De Morgan: invert each input first, $(A' + B')' = A \cdot B$
5. Universal Gate — d) Any gate can be built using it

■ Part C – Activity Questions

1. 2-input NOR truth table: $00 \rightarrow 1, 01 \rightarrow 0, 10 \rightarrow 0, 11 \rightarrow 0$.
2. NOR as NOT: tie both inputs together $\rightarrow Y = (A+A)' = A'$. Truth table: $0 \rightarrow 1, 1 \rightarrow 0$.
3. AND using only NOR gates: invert A (NOR with tied inputs) $\rightarrow A'$; invert B $\rightarrow B'$; then NOR the two results: $(A' + B')' = A \cdot B$. Total 3 NOR gates.